The Single Composition Theorem for the Cross-Domain Unification

A categorical replacement for the seven-rung ladder

The source transcript composes six construction arcs by a seven-rung ladder of category-theoretic vocabulary items. Each rung is a name; no rung is supplied with a composition theorem. The ladder is a sequence of names rather than a sequence of constructions.

This document constructs the missing composition theorem. Each arc is an optic in Optic(C), the monoidal category of optics over a category C with finite limits. The seven-fold composition is well-defined, associative, and unital, by the monoidal structure of Optic(C). The fixed point of the iterated composition, when it exists, is the unification object.

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Theoretical construction (research target)
Companion to: surviving_findings_report.pdf

SECTION 1

Motivation and Setting

The source transcript's seven-rung ladder is a sequence of category-theoretic vocabulary items without composition theorems. The single composition theorem provides the missing construction.

The source transcript composes six construction arcs by a seven-rung ladder of category-theoretic vocabulary items: profunctor, span, optic, lens, dependent type, natural transformation, and so on. Each rung is introduced by name. No rung is supplied with a composition theorem that takes the previous rung's output as input and produces the next rung's input as output. The ladder is therefore a sequence of names rather than a sequence of constructions.

This document constructs the missing composition theorem. The setting is the monoidal category Optic(C) of optics over a category C with finite limits, due to Riley (2018) and Brunerie et al (2020). An optic is a bidirectional state-passing computation: a forward component (the encoding) and a backward component (the decoding), with a residual that carries information between the two. Optic(C) is monoidal under optic composition: the composite of two optics is well-defined, associative, and unital, with the residual of the composite being the product of the residuals of the components.

Each arc of the source transcript is naturally an optic in Optic(C). The RAF arc is an optic whose forward component is the rate-distortion encoder and whose backward component is the rate-distortion decoder, with the residual being the distortion information. The RPSI arc is an optic whose forward component is the predictor and whose backward component is the measurement update, with the residual being the prediction error. The IFS arc is an optic whose forward component is the Hutchinson operator and whose backward component is the deconvolution, with the residual being the contraction factor. The other arcs (Noether, perturbation, WCIG, n=3 Fisher-Rao) admit analogous optic decompositions.

SECTION 2

The Optic Category

Definition of Optic(C) and its monoidal structure. The optic composition operation is the formal replacement for the source transcript's rung-by-rung ladder.

2.1 Definition (Optic)

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An optic over a category C with finite limits is a quadruple (S, A, M, N)
where S is the forward state type, A is the forward action type, M is the
backward state type, and N is the backward action type, together with a
forward morphism fwd : S × Res → A and a backward morphism bwd : S × A × Res
→ M × Res where Res is the residual type. The residual carries information
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from the forward pass to the backward pass.
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In the simpler case where the residual is trivial (Res is the terminal object), the optic reduces to a lens (the symmetric special case). In the case where the residual is the only state and the forward state is trivial, the optic reduces to a coalgebra with residual, which is the structure identified in Section 8 of the surviving findings report as the form of the Blahut-Arimoto operator.

2.2 Definition (Optic composition)

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Given two optics O1 = (S1, A1, M1, N1, fwd1, bwd1) with residual Res1 O2 =
(S2, A2, M2, N2, fwd2, bwd2) with residual Res2 with the compatibility
condition A1 = M2 (the forward action of O1 is the backward state of O2),
the composite optic O2 ∘ O1 = (S1, A2, M1, N2, fwd, bwd) has residual Res1 ×
Res2 and is defined by fwd(s, (r1, r2)) = fwd2(fwd1(s, r1), r2) bwd(s, a,
(r1, r2)) = let (m, r1') = bwd1(s, a, r1) let (n, r2') = bwd2(fwd1(s, r1),
a, r2) in ((n, m), (r1', r2')) The composite is well-defined by the
universal property of the product Res1 × Res2 in C.
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The composite optic O2 ⊠ O1 takes a state in S1, applies O1's forward component to produce an action in A1 (= M2), applies O2's forward component to that action to produce an action in A2, then runs the backward components in reverse order: O2's backward produces N2 and an updated residual r2', then O1's backward produces M1 (= N2-input) and an updated residual r1'. The residual of the composite is the product Res1 × Res2, updated in place.

2.3 Proposition (Monoidal structure of Optic(C))

Optic(C) is a monoidal category under the composition operation of Definition 2.2. The tensor product is the optic composition O2 ⊠ O1; the unit object is the identity optic Id = (S, S, S, S, snd, snd) with trivial residual. The associativity and unitality axioms follow from the universal property of the product Res1 × Res2 in C and the terminality of the unit residual in C.

Proof sketch. The associativity axiom (O3 ⊠ (O2 ⊠ O1)) = ((O3 ⊠ O2) ⊠ O1) follows from the associativity of the product in C: the residual of both sides is (Res1 × Res2) × Res3, which is canonically isomorphic to Res1 × (Res2 × Res3). The forward and backward morphisms agree modulo this canonical isomorphism. The unitality axioms (Id ⊠ O = O and O ⊠ Id = O) follow from the terminality of the unit residual: the product of any residual with the terminal object is canonically isomorphic to the original residual. QED.

SECTION 3

The Seven Arcs as Optics

Each arc of the source transcript is an optic in Optic(C). The table below gives the forward component, backward component, and residual of each arc's optic.

Arc	Forward component (encoding)	Backward component (decoding)	Residual
1. RAF rate-distortion	dist_D encoder: x → (p, x⊠) with d(x, x⊠) ⩽ D	RAF transition: (p, x⊠) → x' via deterministic delta	Distortion d(x, x⊠) (Bregman divergence)
2. RPSI consciousness	Predictor: ρ_in → ρ⊠_out (CPTP channel)	Measurement update: ρ_out → ρ⊠_in (Holevo)	Prediction error I(ρ_out; ρ⊠_in)
3. IFS fractals	Hutchinson operator: K → ∪ f_i(K)	Deconvolution: attractor → component maps	Contraction factor c_i of f_i
4. Noether symmetry	Symmetry map: g_t on dual affine coordinate	Conserved-current extraction: ∂L/∂g_t = 0	Bregman divergence D_phi (affine invariant)
5. Perturbation theory	BA iteration: (p, q) → (p', q')	Derivative of BA: (δp, δq) → (δp', δq')	Distortion derivative ∂d/∂p
6. WCIG	Wasserstein-Categorical embedding: p → sqrt(p)	Inverse embedding: sqrt(q) → q	Fisher-Rao metric g_FR
7. n=3 Fisher-Rao geometry	Connection ∇ on CO(2)-bundle over Theta	Curvature F_∇: Theta × Theta → Lie(CO(2))	Viability margin E and maintenance graph Γ

The table above assigns to each arc a forward component, a backward component, and a residual. The forward components are the encoding operations of each arc; the backward components are the decoding operations; the residuals are the information that flows from the forward pass to the backward pass. The compatibility condition (forward action of arc i equals backward state of arc i+1) is satisfied by construction: each arc's forward action is the next arc's backward state, by the chain structure of the unification.

Two arcs deserve comment. The RPSI arc's forward component is a CPTP channel, not a deterministic function, because the RPSI self-reference paradox requires the quantum lift of Section 9 of the surviving findings report. The n=3 Fisher-Rao arc's residual is the viability margin E and the maintenance graph Gamma, which together encode the autopoietic structure of Section 6 of the surviving findings report. These two arcs carry the heaviest theoretical commitments; the other five arcs are classical and deterministic.

SECTION 4

The Single Composition Theorem

The seven-fold composition of the seven optics is well-defined, associative, and unital. The fixed point of the iterated composition is the unification object.

4.1 Theorem (Single Composition)

THEOREM

Let C be a category with finite limits. Let O_1, O_2, ..., O_7 be the seven optics in Optic(C) defined in Section 3, satisfying the compatibility condition A_i = M_{i+1} for i = 1, ..., 6. Then the seven-fold composition T = O_7 ⊠ O_6 ⊠ ... ⊠ O_1 is well-defined in Optic(C). T is an endofunctor on Optic(C) (in fact, an endomorphism in the monoidal category). The composition is associative and unital. The residual of T is the product Res_1 × Res_2 × ... × Res_7, with the canonical associativity and unitality isomorphisms.

4.2 Proof

By Proposition 2.3, Optic(C) is a monoidal category under the optic composition of Definition 2.2. The seven-fold composition T = O_7 ⊠ O_6 ⊠ ... ⊠ O_1 is therefore well-defined as the iterated monoidal product. The associativity axiom of the monoidal category gives (O_7 ⊠ O_6) ⊠ (O_5 ⊠ O_4) ⊠ (O_3 ⊠ O_2) ⊠ O_1 = O_7 ⊠ (O_6 ⊠ O_5) ⊠ (O_4 ⊠ O_3) ⊠ (O_2 ⊠ O_1) = ... = the canonical seven-fold product, by the associativity isomorphism. All parenthesizations are equal modulo the canonical associativity isomorphisms.

The unitality axiom gives T ⊠ Id = T and Id ⊠ T = T, where Id is the identity optic of Section 2.3. The residual of T is Res_T = Res_1 × Res_2 × ... × Res_7, with the canonical associativity and unitality isomorphisms from C (which is a category with finite limits, so finite products exist and are well-defined up to canonical isomorphism).

The forward and backward components of T are obtained by iterated application of Definition 2.2. The forward component of T is the function that takes a state in S_1 and a residual in Res_T, applies fwd_1 to produce an action in A_1 = M_2, applies fwd_2 to produce an action in A_2 = M_3, and so on, until fwd_7 produces an action in A_7. The backward component of T runs the bwd_i in reverse order, producing the final state in M_1 and updating all residuals. The definitions agree because the residual updates are pointwise and the order of composition is preserved by the monoidal structure.

QED.

4.3 Corollary (Unification object)

COROLLARY

Under the conditions of Theorem 4.1, suppose further that the iterated composition T: Optic(C) → Optic(C) has a fixed point O* with T(O*) = O*. Then O* is the unification object of the cross-domain unification. The forward component of O* encodes the entire RAF → RPSI → IFS → Noether → Perturbation → WCIG → n=3 Fisher-Rao chain in a single encoding; the backward component decodes the entire chain in a single decoding; the residual of O* is the product of all seven residuals, encoding the complete information flow of the chain.

4.4 Sufficient condition for fixed point existence

A sufficient condition for the existence of the fixed point O* is the Bregman-regularized contraction of T, in the sense of Section 8.3 of the surviving findings report. Specifically, suppose that T, viewed as an operator on the space of optics over the n=3 Fisher-Rao base (with the Hausdorff metric on the space of compact subsets of the parameter space), is a contraction under Bregman regularization. Then T has a unique fixed point O*, by the Banach contraction theorem.

The Bregman-regularized contraction condition is checkable empirically: compute the iterates T(O), T^2(O), T^3(O), ..., and verify that the Hausdorff distance between successive iterates decreases geometrically. If it does, the contraction holds and the unification object exists. If it does not, the unification object does not exist in this setting, and a different unification construction (or a different category) is required.

SECTION 5

Implications

The single composition theorem converts the source transcript's rhetorical ladder into a rigorous categorical construction. The implications for the cross-domain unification program are direct.

The single composition theorem converts the source transcript's seven-rung ladder from a sequence of names into a sequence of constructions. Each rung of the ladder is now an optic with a defined forward component, backward component, and residual. The composition of consecutive rungs is well-defined by the monoidal structure of Optic(C). The seven-fold composition is well-defined, associative, and unital.

The unification claim of the source transcript is now formally expressible: the unification object is the fixed point of the iterated composition T. The existence of the fixed point is conditional on the Bregman-regularized contraction of T, which is an empirical question. If the contraction holds, the unification object exists and is unique; if it does not, the unification object does not exist in this setting, and a different construction (or a different category) is required.

The theorem does not assert that the unification object exists; it asserts that the question of the unification object's existence is well-defined and checkable. This is the difference between the source transcript's rhetorical ladder and the rigorous categorical construction. The rhetorical ladder asserts a unification by analogy; the rigorous construction asserts a unification by fixed point of a well-defined operator, with a checkable existence condition.

The theorem is falsifiable. The Bregman-regularized contraction of T can be checked empirically: compute the iterates of T on a starting optic, measure the Hausdorff distance between successive iterates, and fit the rate of decrease. Geometric decrease confirms the contraction and the existence of the unification object; non-geometric decrease refutes the contraction and the existence of the unification object in this setting. The check is operational and gives a clear verdict.

The theorem is the formal replacement for the source transcript's unification claim. It is a research target, not an achieved result: the Bregman-regularized contraction must be verified empirically. The verification is a concrete research program: implement the seven optics in a numerical simulation, compute the iterates of T, and measure the Hausdorff distance. This is the open problem of Section 15 of the surviving findings report, now stated as a falsifiable research target rather than a vague aspiration.

SECTION 6

References

Categorical and information-geometric references underpinning the single composition theorem.

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